CiliaTracks

Biological Workflow Protocol

Table of Contents

1 Overview……1
2 Materials……1
3 Microsphere Application……1
4 Capturing Particle Movement……2

1. Overview

CiliaTracks is a comprehensive Python package and dedicated website that enables clinicians and researchers to analyze cilia-driven particle motility and accurately predict PCD from high-speed video microscopy (HSVM) data.

To carry out analysis with CiliaTracks, HSVM data must first be collected through a standardized biological protocol. This workflow involves applying fluorescent microspheres to cultured airway liquid interface (ALI) cells and capturing their movement.

2. Materials

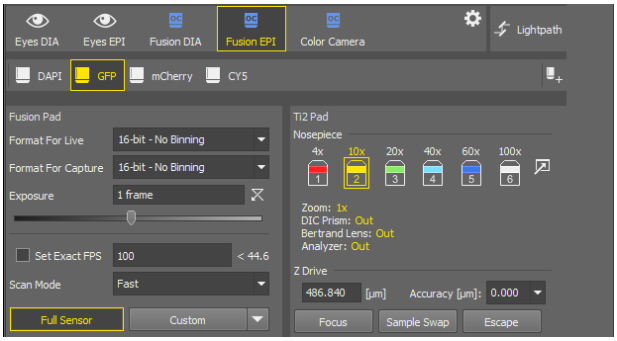
- 1% Solids 1.9 μm diameter green, fluorescent polymer microspheres (Thermo Fisher Scientific, CAT. NO. G0200)
- 2% Nu-Serum (Corning, Product Number 355100) media
- ALI differentiated cultured airway epithelial cells in Transwells (6.5 mm or 12 mm diameter, Product Numbers 3472 & 3462)
- Widefield inverted fluorescence microscope capable of capturing videos at 100 fps

3. Microsphere Application

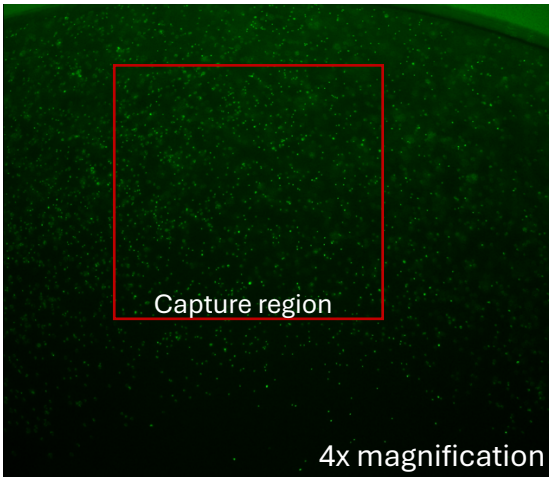
1. Dilute 1% solids, 1.9 μm diameter green, fluorescent polymer microspheres 1:1000 in warm 2% Nu-Serum
    a. Incubate at 37°C for a minimum of 10 minutes
2. Acquire cultured cells of interest (differentiated on Transwell plates)
    a. Gently wash the apical chamber with warm 2% Nu-Serum, lightly rotating for 1 minute
    b. Aspirate media
3. Mix diluted fluorescent microspheres solution and apply beads to surface of inserts slowly, aiming for the center
    a. Add 20 μL for 6.5 mm diameter Transwells, 60 μL for 12 mm
4. Incubate cells for 10 minutes at 37 °C

4. Capturing Particle Movement

1. Turn on microscope, and its environmental chamber, setting it to 37°C
2. Acquire cells from incubator and place in microscope
3. Initialize following microscope settings:
    a. Set to Fusion EPI viewing mode
    b. Set to GFP fluorescence channel
    c. 10x magnification



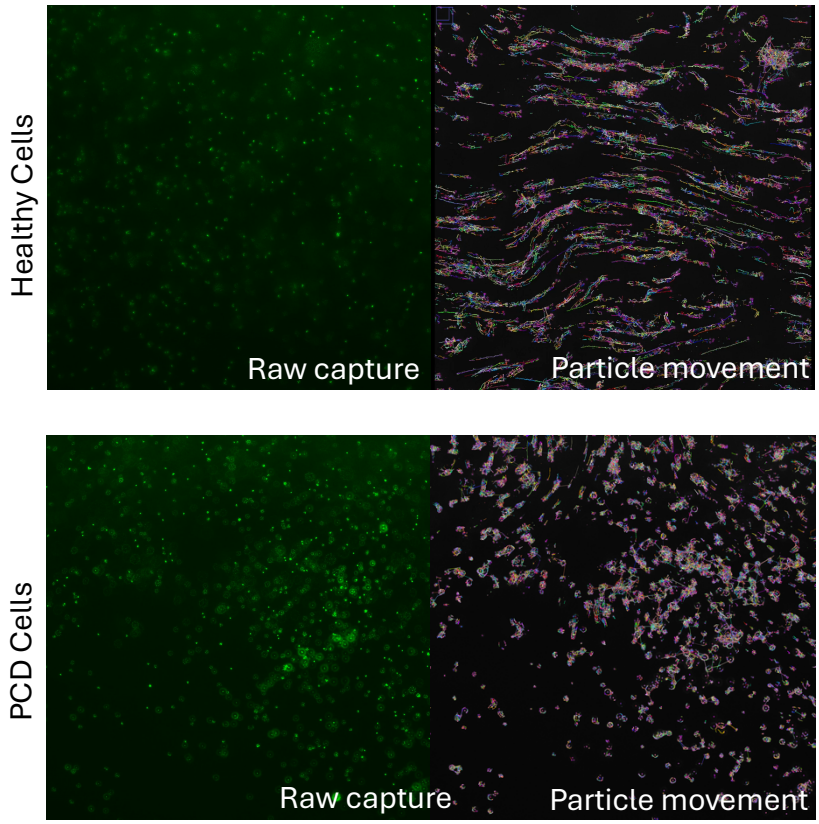
4. Visualize and identify regions on insert with maximum bead movement
    a. Best regions for capture are typically away from the center of insert



5. Once a suitable region is identified, ensure focus and contrast are well adjusted. Capture a 4.45 second video at 100 fps and 10x magnification.
6. Note down the µm/pixel ratio for the specified resolution



7. Save video as .avi file with MJPEG compression if possible
8. Capture a minimum of 3 videos per insert



9. Once finished capturing process, rinse inserts with 2% Nu-Serum three times
10. Use CIliaTracks for subsequent analysis